

COTULLA-LA SALLE COUNTY AP, TX, USA

WMO: 722526

Lat: 28.457N

Lon: 99.218W

Elev: 476

StdP: 14.44

Time Zone: -6.00 (NAC)

Period: 01-19

WBAN: 12947

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	32.1	35.7	11.6	10.1	47.4	16.0	12.6	52.1	24.0	63.3	20.1	61.4	4.4	50	0.440

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	23.9	103.4	73.2	101.7	73.4	99.9	73.6	78.9	89.1	78.2	88.5	77.5	87.9	10.4	150

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
76.8	142.2	81.2	75.9	137.8	80.8	75.2	134.4	80.5	42.7	89.5	41.9	88.8	41.2	87.8	84.4

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 20.5	18.3	16.5	26.2	106.9	3.8	2.3	23.5	108.6	21.3	109.9	19.2	111.2	16.4	112.9		
(5)			23.3	80.8	3.6	1.5	20.7	81.9	18.6	82.8	16.6	83.6	14.0	84.7		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	73.0	55.6	60.3	66.9	74.1	80.4	85.6	86.3	87.4	82.3	74.7	64.6	56.8		
	DBStd	13.31	9.14	9.94	8.47	7.01	5.71	3.27	3.37	2.96	4.70	7.48	8.98	8.80		
	HDD50	115	47	24	5	0	0	0	0	0	0	0	7	32		
	HDD65	1015	315	189	82	13	1	0	0	0	0	17	117	281		
	CDD50	8504	220	312	529	723	941	1067	1126	1161	970	765	446	244		
	CDD65	3929	23	58	141	286	477	617	661	696	521	317	105	27		
	CDH74	46130	241	541	1227	2814	5300	7916	8601	9528	5755	3128	847	232		
	CDH80	24550	54	177	436	1278	2720	4542	4998	5721	3005	1338	240	41		
(14) Wind	WSAvg	8.0	6.7	7.6	8.5	9.2	9.6	9.3	8.9	8.3	7.0	7.2	6.6	6.3		
(15) Precipitation	PrecAvg	21.7	1.1	1.2	1.6	1.7	2.8	2.2	2.3	2.0	2.8	2.2	1.3	1.1		
	PrecMax	37.6	4.4	3.5	4.3	4.9	7.6	6.4	10.7	7.4	7.7	9.2	4.4	4.3		
	PrecMin	9.6	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.2	0.2	0.2	0.0	0.0		
	PrecStd	6.7	1.1	1.2	1.3	1.5	2.1	1.9	2.8	2.2	2.0	2.1	1.3	1.2		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	85.4	91.0	93.0	100.3	102.4	105.6	105.1	105.6	103.0	96.4	90.0	84.5		
		MCWB	63.8	63.5	65.7	70.2	73.7	73.3	73.1	73.5	71.5	73.0	70.0	66.0		
	2%	DB	80.7	85.2	88.7	95.1	99.2	101.9	102.9	103.3	99.5	93.1	85.8	80.1		
		MCWB	62.2	63.9	66.6	69.1	73.8	73.8	73.2	73.5	73.1	72.6	68.4	64.0		
	5%	DB	75.7	81.0	85.0	91.2	96.4	99.7	100.8	101.4	96.7	90.3	82.4	76.1		
		MCWB	60.3	63.8	65.9	70.0	73.5	73.8	73.6	73.7	73.3	71.4	67.4	62.9		
	10%	DB	71.6	76.0	81.2	87.5	92.9	97.3	98.6	99.5	93.7	87.8	79.1	72.2		
		MCWB	59.9	62.3	65.2	68.4	72.9	73.9	73.9	74.0	73.4	71.0	66.1	62.3		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	69.1	71.9	72.6	75.5	79.0	80.2	79.4	79.2	79.6	78.5	74.3	71.0		
		MCDB	76.8	79.2	82.3	87.3	90.6	91.4	89.4	90.4	86.8	85.1	81.5	77.6		
	2%	WB	67.2	69.2	71.0	74.3	77.7	78.7	78.4	78.2	78.1	76.9	72.4	69.1		
		MCDB	74.4	77.6	80.1	85.7	89.2	89.8	88.7	89.3	86.2	84.3	79.8	75.5		
	5%	WB	64.9	67.4	69.7	73.2	76.6	77.8	77.7	77.4	77.3	75.7	70.9	67.0		
		MCDB	71.6	74.8	78.0	84.1	87.8	88.7	88.3	88.6	85.7	83.1	78.1	72.6		
	10%	WB	62.1	65.6	68.6	72.1	75.6	77.0	77.1	76.9	76.5	74.6	69.1	64.3		
		MCDB	68.7	72.5	76.7	82.1	86.3	87.7	87.7	88.2	85.0	82.1	76.3	70.1		
(35) Mean Daily Temperature Range	5% DB	MDBR	22.9	23.1	22.9	23.8	22.8	23.3	22.9	23.9	21.1	22.6	22.4	22.4		
		MCDBR	29.3	28.8	27.7	28.0	26.8	26.7	26.8	26.7	24.7	23.6	24.1	27.7		
		MCWBR	14.4	13.0	10.8	9.0	6.8	5.4	5.5	5.4	6.0	7.5	10.4	14.1		
	5% WB	MCDBR	21.4	21.8	20.3	21.9	21.5	22.4	22.0	23.2	19.6	19.2	19.8	20.4		
		MCWBR	12.5	12.6	9.4	8.2	6.9	5.9	5.5	5.2	5.6	7.9	10.3	12.5		
(40) Clear-Sky Solar Irradiance	taub		0.337	0.342	0.362	0.400	0.430	0.417	0.417	0.415	0.404	0.368	0.354	0.328		
	taud		2.474	2.466	2.424	2.336	2.300	2.392	2.400	2.413	2.396	2.489	2.464	2.515		
	Ebn at Noon		287	295	294	285	275	276	276	277	277	282	278	285		
	Edh at Noon		30	33	36	41	42	38	38	37	37	31	30	28		
(44) All-Sky Solar Radiation	RadAvg		983	1178	1482	1744	1883	2154	2140	2063	1689	1411	1071	885		
	RadStd		121	208	173	177	190	124	172	102	149	180	127	134		

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)		
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+436	
(47) Regional (22 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+207	+173

Nomenclature: See separate page